Converting Units

Name _____ **Date** _____ No calculator Version 1

KEY WORDS convert · metric · length · mass · capacity · area · volume · square units · cubic units

A Length ●○○

ANSWER ON THE SHEET

Remember: 1 cm = 10 mm, 1 m = 100 cm, 1 km = 1,000 m. Multiply to change to a smaller unit. Divide to change to a larger unit.

1. Write the missing number.

a) 4 cm = ___ mm **b)** 3 m = ___ cm **c)** 2 km = ___ m **d)** 70 mm = ___ cm

e) 450 cm = ___ m **f)** 3,500 m = ___ km **g)** 2.5 m = ___ cm **h)** 0.8 km = ___ m

i) 36 mm = ___ cm **j)** 1.2 m = ___ mm **k)** 75 cm = ___ m **l)** 640 m = ___ km

B Mass and capacity ●○○

ANSWER ON THE SHEET

Remember: 1 kg = 1,000 g, 1 tonne (t) = 1,000 kg, 1 L = 1,000 mL.

2. Write the missing number.

a) 3 kg = ___ g **b)** 5,000 g = ___ kg **c)** 2 L = ___ mL **d)** 4,000 mL = ___ L

e) 1.5 kg = ___ g **f)** 750 g = ___ kg **g)** 0.25 L = ___ mL **h)** 3,200 mL = ___ L

i) 4 t = ___ kg **j)** 2,500 kg = ___ t **k)** 0.6 kg = ___ g **l)** 85 mL = ___ L

C Units of area ●●○

ANSWER ON THE SHEET

Watch out: A square centimetre is 10 mm by 10 mm, so 1 cm^2 = 100 mm^2, not 10 mm^2. In the same way 1 m^2 = 10,000 cm^2 and 1 km^2 = 1,000,000 m^2.

3. Write the missing number.

a) 3 cm^2 = ___ mm^2 **b)** 5 m^2 = ___ cm^2 **c)** 2 km^2 = ___ m^2

d) 700 mm^2 = ___ cm^2 **e)** 40,000 cm^2 = ___ m^2 **f)** 0.5 m^2 = ___ cm^2

g) 2.4 cm^2 = ___ mm^2 **h)** 8,500 cm^2 = ___ m^2 **i)** 60 mm^2 = ___ cm^2

D Units of volume ●●●

ANSWER ON THE SHEET

Watch out: A cubic centimetre is 10 mm by 10 mm by 10 mm, so 1 cm^3 = 1,000 mm^3. In the same way 1 m^3 = 1,000,000 cm^3.

4. Write the missing number.

a) 2 cm^3 = ___ mm^3 **b)** 3 m^3 = ___ cm^3 **c)** 5,000 mm^3 = ___ cm^3

d) 4,000,000 cm^3 = ___ m^3 **e)** 0.5 cm^3 = ___ mm^3 **f)** 750 mm^3 = ___ cm^3

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E Volume and capacity ●●○ — ANSWER ON THE SHEET

> **Remember:** Capacity is how much a container holds. 1 mL = 1 cm³, 1 L = 1,000 cm³ and 1 m³ = 1,000 L.

5. Write the missing number.

a) 250 cm³ = ___ mL

b) 3 L = ___ cm³

c) 5,000 cm³ = ___ L

d) 2 m³ = ___ L

e) 4,000 L = ___ m³

f) 1.5 L = ___ cm³

g) 750 cm³ = ___ L

h) 0.5 m³ = ___ L

i) 600 mL = ___ cm³

F Problems ●●● — WORK IN YOUR BOOK

6. A bottle holds 1.5 litres of juice. How many 250 mL glasses can be filled from it?

7. A piece of wood is 3.2 m long. It is cut into pieces that are 80 cm long. How many pieces are there?

8. A fish tank is 20 cm long, 15 cm wide and 10 cm deep. How many litres of water does it hold when it is full?

9. A rug has an area of 2 m². A tile has an area of 500 cm². How many tiles would cover the same area as the rug?

★ Extension — FOR EARLY FINISHERS

10. A cube has edges 50 cm long. Find its volume in cm³, then write its capacity in litres.

11. Write 0.03 km² in square metres.

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Cell Tuition
1-1 PRIVATE TUTORS

YEAR 8 · MEASUREMENT

Converting Units

Name _____ **Date** _____ No calculator Version 2

KEY WORDS convert · metric · length · mass · capacity · area · volume · square units · cubic units

A Length ●○○

ANSWER ON THE SHEET

Remember: 1 cm = 10 mm, 1 m = 100 cm, 1 km = 1,000 m. Multiply to change to a smaller unit. Divide to change to a larger unit.

1. Write the missing number.

a) 6 cm = ___ mm **b)** 5 m = ___ cm **c)** 4 km = ___ m **d)** 90 mm = ___ cm

e) 250 cm = ___ m **f)** 1,500 m = ___ km **g)** 3.5 m = ___ cm **h)** 0.6 km = ___ m

i) 48 mm = ___ cm **j)** 1.5 m = ___ mm **k)** 45 cm = ___ m **l)** 820 m = ___ km

B Mass and capacity ●○○

ANSWER ON THE SHEET

Remember: 1 kg = 1,000 g, 1 tonne (t) = 1,000 kg, 1 L = 1,000 mL.

2. Write the missing number.

a) 4 kg = ___ g **b)** 7,000 g = ___ kg **c)** 3 L = ___ mL **d)** 6,000 mL = ___ L

e) 2.5 kg = ___ g **f)** 250 g = ___ kg **g)** 0.75 L = ___ mL **h)** 1,800 mL = ___ L

i) 6 t = ___ kg **j)** 3,500 kg = ___ t **k)** 0.4 kg = ___ g **l)** 65 mL = ___ L

C Units of area ●●○

ANSWER ON THE SHEET

Watch out: A square centimetre is 10 mm by 10 mm, so 1 cm^2 = 100 mm^2, not 10 mm^2. In the same way 1 m^2 = 10,000 cm^2 and 1 km^2 = 1,000,000 m^2.

3. Write the missing number.

a) 4 cm^2 = ___ mm^2 **b)** 3 m^2 = ___ cm^2 **c)** 5 km^2 = ___ m^2

d) 900 mm^2 = ___ cm^2 **e)** 60,000 cm^2 = ___ m^2 **f)** 0.2 m^2 = ___ cm^2

g) 3.6 cm^2 = ___ mm^2 **h)** 7,500 cm^2 = ___ m^2 **i)** 80 mm^2 = ___ cm^2

D Units of volume ●●●

ANSWER ON THE SHEET

Watch out: A cubic centimetre is 10 mm by 10 mm by 10 mm, so 1 cm^3 = 1,000 mm^3. In the same way 1 m^3 = 1,000,000 cm^3.

4. Write the missing number.

a) 4 cm^3 = ___ mm^3 **b)** 2 m^3 = ___ cm^3 **c)** 7,000 mm^3 = ___ cm^3

d) 6,000,000 cm^3 = ___ m^3 **e)** 0.2 cm^3 = ___ mm^3 **f)** 250 mm^3 = ___ cm^3

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E Volume and capacity ●●○ — ANSWER ON THE SHEET

> **Remember:** Capacity is how much a container holds. 1 mL = 1 cm³, 1 L = 1,000 cm³ and 1 m³ = 1,000 L.

5. Write the missing number.

a) 350 cm³ = ___ mL

b) 4 L = ___ cm³

c) 7,000 cm³ = ___ L

d) 3 m³ = ___ L

e) 6,000 L = ___ m³

f) 2.5 L = ___ cm³

g) 250 cm³ = ___ L

h) 0.2 m³ = ___ L

i) 400 mL = ___ cm³

F Problems ●●● — WORK IN YOUR BOOK

6. A bottle holds 2 litres of juice. How many 250 mL glasses can be filled from it?

7. A piece of wood is 2.4 m long. It is cut into pieces that are 60 cm long. How many pieces are there?

8. A fish tank is 30 cm long, 20 cm wide and 10 cm deep. How many litres of water does it hold when it is full?

9. A rug has an area of 3 m². A tile has an area of 500 cm². How many tiles would cover the same area as the rug?

★ Extension — FOR EARLY FINISHERS

10. A cube has edges 40 cm long. Find its volume in cm³, then write its capacity in litres.

11. Write 0.07 km² in square metres.

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